Software Requirements Engineering

Quiz Answer Key — Mid Quiz 1 & Mid Quiz 3

Answers cross-checked against Wiegers & Beatty, Software Requirements (3rd ed.) and standard requirements-engineering references. Flagged items are genuinely source-dependent — see the note at the end.

Quiz 1 — Circle the Correct Answer

10 multiple-choice questions

Q1. Requirement engineering process includes which of these steps.

A) Feasibility study B) Requirement Gathering
C) Software Requirement specification & Validation D) All mentioned above

Answer: (D) All mentioned above

Why: Requirements engineering covers feasibility study, elicitation/gathering, specification, and validation - so all three earlier options are part of the same overall process.

Q2. Which of the following is considered to be external quality attributes?

A) Modifiability B) Portability
C) Reusability D) Reliability

Answer: (D) Reliability

Why: External attributes are the ones a user can observe just by running the system, like reliability, efficiency, usability, and integrity. Modifiability, portability, and reusability matter mainly to developers, so they're internal attributes.

Q3. Which of the following is considered to be internal quality attributes?

A) Availability B) Integrity
C) Interoperability D) Verifiability

Answer: (D) Verifiability

Why: Verifiability - how easy it is to test and confirm correctness - is a pure developer concern, which is the cleanest fit for "internal." Availability and integrity are usually grouped with reliability and security, which are external/user-facing.

Double-check: Depending on which quality model your course slides use, Interoperability (C) is sometimes also classified as internal. If your notes list it that way, that's a legitimate alternate framework, not an error - worth confirming against your exact lecture source.

Q4. The system shall be developed using open source tools and shall run on Linux OS. This is:

A) Functional requirement B) Non-Functional requirements
C) Inverse requirement D) Constraints

Answer: (D) Constraints

Why: This dictates a specific technology and platform rather than describing behavior or a quality - that makes it a design/implementation constraint.

Q5. The system shall not disclose the salary information of the staff of a school. This is:

A) Functional requirement B) Non-Functional requirements
C) Inverse requirement D) Constraints

Answer: (B) Non-Functional requirements

Why: This is a confidentiality/security expectation - a quality attribute - rather than a description of a specific feature, so it's non-functional.

Q6. Login into a system requires user name and password, is a

A) Functional requirement B) Non-Functional requirement
C) Inverse requirement D) System Requirement

Answer: (A) Functional requirement

Why: It describes an explicit behavior the system must perform (authenticate with a username and password), which is exactly what a functional requirement captures.

Q7. Any unauthorized access to the customer data shall be prevented, is a

A) Functional requirement B) Non-Functional requirements
C) Inverse requirement D) Constraints

Answer: (B) Non-Functional requirements

Why: Preventing unauthorized access is a security expectation, so this is a non-functional (quality) requirement rather than a specific feature description.

Q8. The requirements that describes the efficiency of a system, is a

A) Functional requirement B) Non-Functional requirements
C) Inverse requirement D) Constraints

Answer: (B) Non-Functional requirements

Why: Efficiency (performance) is one of the classic quality attributes, so it falls under non-functional requirements.

Q9. The requirements that describes the testability of a system, is a

A) Functional requirement B) Quality requirements
C) Business requirement D) System requirements

Answer: (B) Quality requirements

Why: Testability is a quality characteristic of the system, so "Quality requirements" is the best fit here.

Double-check: Your printed sheet has a typo - both of the first two options are labeled "(B)". I've treated the first as (A) Functional requirement and kept (B) Quality requirements as the answer, since that's clearly the intended pairing.

Q10. If a sentence gives two different meaning to two different readers, then we call it ___.

A) Inconsistencies B) Ambiguous
C) Redundancies D) Confliction

Answer: (B) Ambiguous

Why: A statement that different readers interpret differently is the textbook definition of ambiguity.

Descriptive Question: Describe the Roles and Responsibilities of Customers to collect the requirements in best way.

This maps to Wiegers & Beatty's "Requirements Bill of Responsibilities for Software Customers" - what customers owe the analyst/development team in return for a good requirements process:

- Educate analysts and developers about your business - they can't understand your domain without your help.
- Dedicate real time to provide requirements and answer clarifying questions, rather than rushing through it.
- Be specific and precise about what you need instead of describing vague wishes.
- Make timely decisions whenever the team needs an answer or a choice made.
- Respect the analyst's or developer's assessment of cost and feasibility, instead of assuming everything is easy to build.
- Set clear priorities among your requirements - not everything can be "most important."
- Review requirements documents, models, and prototypes carefully and give real, timely feedback.
- Communicate changes to your requirements as soon as you know about them, not at the last minute.
- Follow the agreed change-control process for requesting modifications, rather than routing around it informally.
- Respect the processes the analysts and developers use to do their job.

Quiz 3 — True / False

10 statements

1. A wireframe is a particular approach to throwaway prototyping commonly used for custom user interface design and website design.

Answer: True

Why: Wireframes are a classic low-fidelity, throwaway prototyping technique used specifically to work out UI layout and website structure before real design or coding begins.

2. Prioritization is a critical strategy for projects that develop products through a series of fixed-schedule time boxes.

Answer: True

Why: When delivery happens in fixed timeboxes (like agile sprints), not everything can fit in every box, so prioritization is what decides what makes the cut each time.

3. The benefits of effective requirements reuse include faster delivery.

Answer: True

Why: Reusing requirements (templates, patterns, prior specifications) cuts down on rework, which is a well-documented benefit alongside lower cost, consistency, and fewer defects.

4. The best estimation processes acknowledge the early uncertainty and ongoing volatility of scope.

Answer: True

Why: This is the "cone of uncertainty" idea - good estimating practices explicitly account for estimates being least reliable early on, and for scope continuing to shift as the project proceeds.

5. The purpose of prototyping is clarify, complete, and validate requirements.

Answer: True

Why: Clarifying vague requirements, filling in missing ones, and validating them with stakeholders are the three standard purposes of prototyping in requirements work.

6. Prototyping helps in exploring design alternatives.

Answer: True

Why: Prototypes are a standard, low-cost way to try out and compare competing UI or design options with real users before committing to one.

7. Prototyping creates a subset that will grow into the ultimate product.

Answer: False

Why: This describes evolutionary prototyping specifically, not prototyping in general. A throwaway prototype - the more common approach for clarifying requirements - is built and then discarded, not grown into the final system.

Double-check: If your course material treats "prototyping" as inherently evolutionary, your instructor may have keyed this True instead - worth a quick check, since this is really a definitional choice.

8. Electronic prototyping is a cheap, fast, and low-tech way to explore how a portion of an implemented system might look.

Answer: False

Why: That description (cheap, fast, low-tech) actually matches paper prototyping. Electronic prototypes use software tools, so they take more setup even though they end up looking more realistic.

9. A mock-up performs many useful work, although it looks as if it shouldn't.

Answer: True

Why: A mock-up isn't functional and looks like "just a picture," but it still does real work - catching usability problems, prompting stakeholder feedback, and validating the design - all before a line of code is written.

10. A not so effective way to reuse by reference is to store requirements in a requirements management tool.

Answer: False

Why: Storing and linking requirements inside a requirements management (RM) tool is usually considered one of the more effective ways to reuse by reference, especially compared with copying and pasting text around, which drifts out of sync over time.

Descriptive Question: Describe the approaches we can follow to Elicit requirements. Describe the Traps we should avoid to choose Product Champion.

Requirements elicitation approaches

- Interviews - one-on-one conversations for in-depth understanding of a stakeholder's needs.
- Workshops (e.g. JAD sessions) - structured group sessions where multiple stakeholders work out requirements together in real time.
- Focus groups - guided discussions with a representative group of users to gather reactions and opinions.
- Observation - watching users do their actual jobs, since people don't always think to mention things they do automatically.
- Questionnaires and surveys - useful when stakeholders are numerous or spread across locations.
- System interface analysis - understanding what the new system needs to exchange with other systems.
- User interface analysis - studying existing or competitor interfaces for ideas and pitfalls to avoid.
- Document analysis - mining existing manuals, forms, business rules, and older system specifications.
- Prototyping - showing stakeholders something tangible often surfaces requirements they couldn't describe from scratch.

Product Champion traps to avoid

- Letting managers override a properly authorized champion's decisions at the last minute - it undermines the champion and frustrates the users they represent.
- A champion who forgets they're representing a whole user class and pushes only their own personal preferences.
- A champion with no real vision of the system, who simply defers to whatever the analyst proposes instead of contributing genuine input.

- A senior, busy user handing the role to a junior colleague but continuing to backseat-drive decisions from the sidelines.
- Someone who claims to speak for a user group they don't actually belong to.
- Expecting a single champion to represent a very diverse user base alone - larger or more varied user classes usually need more than one champion.

A couple of things worth double-checking

- Quiz 1, Question 3: marked (D) Verifiability, but (C) Interoperability is sometimes also classified as an internal quality attribute depending on the model your course uses.
- Quiz 3, Question 7: marked False because the statement describes evolutionary prototyping specifically, not prototyping in general - but this hinges on how your course defines “prototyping.”
- Quiz 1, Question 9: the printed sheet has a typo (two options both labeled B) - answered based on the intended second option, “Quality requirements.”